



MANUEL S. ENVERGA UNIVERSITY FOUNDATION
COLLEGE OF COMPUTING & MULTIMEDIA STUDIES
CHED CENTER OF DEVELOPMENT IN INFORMATION TECHNOLOGY

ITWM106

Mobile Application Development 3 (cross-platform)

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Custom Components

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Custom Components

- You can nest components inside each other to create new components.

Custom Components p.1

```

import React from 'react';
import { Text, View } from 'react-native';

const StudentCard = () => {
  return (
    <View>
      <Text>Juan Dela Cruz</Text>
      <Text>BS in Information Technology</Text>
    </View>
  );
}
  
```

Custom Components p.2

```
const StudentList = () => {
  return (
    <View>
      <Text>Welcome!</Text>
      <StudentCard />
      < StudentCard />
      < StudentCard />
    </View>
  );
}
```

export default PersonList;



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Custom Components

- Any component that renders other components is a **parent** component.
- In the example, PersonList is the **parent** component and each Person is a **child** component.



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Props and State

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Props

- Props let you customize React components.
- Most components can be customized when they are created, with different parameters.
- These creation parameters are called props.
- Props is short for “properties”.



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Props Example p.1

```
import React from 'react';
import { Text, View } from 'react-native';

const StudentCard = (props) => {
  return (
    <View style="alignItems: 'center'">
      <Text>Hello { props.name }</Text>
    </View>
  );
}
```



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Props Example p.1

```
const StudentList = () => {
  return (
    <View>
      <Text>Welcome!</Text>
      <StudentCard name='Juan' />
      <StudentCard name='Pedro' />
      <StudentCard name='Marites' />
    </View>
  );
}

export default StudentList;
```



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Props

- Most of React Native's Core Components can be customized with props, too.
- For example, when using Image, you pass it a prop named **source** to define what image it shows



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Props

```
import React from 'react';
import { Text, View, Image } from 'react-native';

const CatApp = () => {
  return (
    <View>
      <Image
        source={{uri: "https://reactnative.dev/docs/assets/p_cat1.png"}}
        style={{width: 200, height: 200}}
      />
      <Text>Hello, I am your cat!</Text>
    </View>
  );
}
export default CatApp;
```



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Props

- Image has many different props, including style, which accepts a JS object of design and layout related property-value pairs
- You can build many things with props and the Core Components Text, Image, and View!
- But to build something interactive, you'll need state.

State

- While you can think of props as **arguments** you use to configure how components render, **state** is like a component's **personal data storage**.
- State is useful for **handling data** that **changes over time** or that comes from **user interaction**.
- State allows React Native components to re-render and update their user interfaces in response to user interactions, network requests, or other events. Stateful components are often referred to as "stateful"
- State gives your components **memory**!

State in Function Component

```
import React, { useState } from "react";
import { Text, View } from "react-native";

const Counter = () => {
  const [num, setNumber] = useState(0);

  return (
    <View>
      <Text style={{ fontSize: 40 }}>
        The number is { num }
      </Text>
    </View>
  );
}

export default Counter;
```



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State in Function Component

- We declare the component's state by calling **useState** inside its function.
- In the example, **useState** creates a "num" state variable
- You can use useState to track any kind of data: strings, numbers, Booleans, arrays, objects.
- For example, you can set the name of a person with `const [name, setName] = useState('Juan')`



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State in Function Component

- Calling `useState` does two things:
 - it creates a "state variable" with an initial value—in this case the state variable is "num" and its initial value is 0
 - it creates a function to set that state variable's value—`setNumber`
- It doesn't matter what names you use. But it can be handy to think of the pattern as [`<getter>`, `<setter>`] = `useState(<initialValue>)`.

State in Function Component

- To set individual values inside the state object, use the function `setNumber`
- You can use `setNumber` inside a button to change the value of the state when the button is pressed

```
<Button title="Update" onPress={() => { setNumber(1); }}/>
```

State vs. Props

- The state is mutable while props are immutable.
- This means that state can be updated in the future while props cannot be updated.
- As a general rule, use props to configure a component when it renders.
- Use state to keep track of any component data that you expect to change over time.

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